

Backend Assignment

Token Insight & Analytics API

Deliverables

- GitHub repo with source code
 - README with setup instructions (Docker preferred)
 - Running backend with required APIs
 - Postman collection (optional but bonus)
 - Good to include: Unit tests.
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Tech Guidelines

- Backend: Node.js/Express **or** Python/FastAPI/Flask
 - Database: Not required (in-memory fine) — SQLite/Postgres optional
 - Market data source: [CoinGecko API](#) (free, no key required)
 - AI: Any free/hosted model (Hugging Face, OpenAI free trial, local llama.cpp).
 - **Do not commit API keys** — use .env + .env.example.
 - Document how to set up AI in README.
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Tasks

1. Token Insight API

Combine **fetching**, **prompting**, and **AI integration** into one endpoint.

Endpoint:

POST /api/token/:id/insight

Flow:

1. Fetch token metadata & market data from CoinGecko (/coins/{id} + optional /market_chart).
2. Build a structured prompt

3. Call chosen AI model with **prompt**.
4. Parse & validate AI response (must be valid JSON).
5. Return combined output.

Request (optional body):

```
{  
  "vs_currency": "usd",  
  "history_days": 30  
}
```

Response (example):

```
{  
  "source": "coingecko",  
  "token": {  
    "id": "chainlink",  
    "symbol": "link",  
    "name": "Chainlink",  
    "market_data": {  
      "current_price_usd": 7.23,  
      "market_cap_usd": 3500000000,  
      "total_volume_usd": 120000000,  
      "price_change_percentage_24h": -1.2  
    }  
  },  
  "insight": {  
    "reasoning": "Generic market comment",  
    "sentiment": "Neutral"  
  },  
}
```

```
"model": { "provider": "openai", "model": "gpt-4o-mini" }  
}
```

2. HyperLiquid Wallet Daily PnL API

You are required to build a backend service that fetches a wallet's daily PnL (Profit and Loss) from **HyperLiquid**.

- Input: Wallet address + date range (start, end).
- Output: Daily realized/unrealized PnL, fees, funding.

Key requirements:

1. API: GET /api/hyperliquid/:wallet/pnl?start=YYYY-MM-DD&end=YYYY-MM-DD
2. Fetch wallet activity from HyperLiquid (trades, positions, funding, fees).
3. Calculate daily:
 - **Realized PnL** → closed trades
 - **Unrealized PnL** → open positions marked to daily close price
 - **Fees & Funding** → charged on that day
 - **Net PnL** = realized + unrealized – fees + funding
4. Return results as JSON (see prototype below).
5. Handle invalid wallet, missing data, and API errors gracefully.

Response (example)

```
{  
  "wallet": "0xabc123...",  
  "start": "2025-08-01",  
  "end": "2025-08-03",  
  "daily": [  
    {  
      "date": "2025-08-01",
```

```
"realized_pnl_usd": 120.5,
"unrealized_pnl_usd": -15.3,
"fees_usd": 2.1,
"funding_usd": -0.5,
"net_pnl_usd": 102.6,
"equity_usd": 10102.6
},
{
  "date": "2025-08-02",
  "realized_pnl_usd": 0,
  "unrealized_pnl_usd": -10.0,
  "fees_usd": 1.2,
  "funding_usd": -0.3,
  "net_pnl_usd": -11.5,
  "equity_usd": 10091.1
}
],
"summary": {
  "total_realized_usd": 120.5,
  "total_unrealized_usd": -25.3,
  "total_fees_usd": 3.3,
  "total_funding_usd": -0.8,
  "net_pnl_usd": 91.1
},
"diagnostics": {
  "data_source": "hyperliquid_api",
```

"last_api_call": "2025-09-22T12:00:00Z",

"notes": "PnL calculated using daily close prices"

}

}